

Aditi Sharma

Email: aditi.sharma.bioinfo@gmail.com

G-Scholar: [Profile](#) · Github: [Sharmaditi29](#) · LinkedIn: [aditisharma29](#)

Research Summary: Computational biologist with experience in bioinformatics, biological data analysis, machine learning, deep learning and computational method development.

Experience

Thermo Fisher Scientific | *Researcher*

DEC 2021 - NOV 2024

Scientific software development

Bremen, Germany

Integrated the open-source NuXL node into Proteome Discoverer 3.1 using C# for RNA/DNA-peptide cross-link analysis. Extended result tables with protein accessions, confidence scores, PSM summaries and component mass breakdowns.

Technical University of Munich | *PhD Student*

DEC 2021 - OCT 2023

Nucleic-acid-peptide cross-link data analysis

Munich, Germany

Curated, quality-checked and harmonized public and partner datasets. Prepared data in Python and adapted the Prosit neural-network model for spectrum prediction using Slurm. This appointment overlapped with Thermo Fisher Scientific.

ICGEB | *Junior Researcher*

JUL 2020 - JUN 2021

Computational genomics and proteomics

New Delhi, India

Built a hybrid genome-assembly and annotation workflow using Illumina and Oxford Nanopore data on HPC systems. Applied CNN-LSTM models to DIA-MS data for de novo peptide sequencing and functional annotation.

Education

Indraprastha Institute of Information Technology Delhi

JUN 2017 - JUL 2019

M.Tech. in Computational Biology

New Delhi, India

Supervisor: Prof. Ganesh Bagler

Thesis: Prediction of Sweetness using Machine Learning Models.

Jaypee University of Information Technology

JUN 2012 - JUL 2016

B.Tech. in Bioinformatics

Solan, India

Supervisor: Late Dr. Jayashree Ramana

Thesis: Meta-analysis of Brain and CNS Cancer microarray datasets.

Selected Publications

- **Machine Learning Models to Predict Sweetness of Molecules**
M. Goel, [A. Sharma](#), A. S. Chilwal, S. Kumari, A. Kumar, G. Bagler
Computers in Biology and Medicine, 2023 — [Publication](#)
- **Meta-Analysis of Brain and Central Nervous System Microarray Datasets**
I. Khandelwal, [A. Sharma](#), J. Ramana
International Journal for Computational Biology, 2018

Other Publications

- **Nature-Inspired Computation in Data Mining**
[A. Sharma](#)
Wiley, 2021 (Book chapter) — [Publication](#)
- **A Repository of Web-Based Bioinformatics Resources Developed in India**
A. Agarwal, P. Agrawal, [A. Sharma](#), V. Kumar, C. Mugdal, G. P. S. Raghava
bioRxiv, 2020 (Preprint) — [Publication](#)
- **Bioinformatics Database Resources**
I. Khandelwal, [A. Sharma](#), P. K. Agrawal, R. Shrivastava
IGI Global, 2017 (Book chapter) — [Publication](#)

Poster Presentations

An end-to-end analysis and visualization workflow for nucleic acids crosslinking data Academic Mindtrek, Finland	OCT 2023
A Protein-Nucleic Acid Cross-Linking Node for Proteome Discoverer 2.5 EuPA (European Proteomics Association)	APR 2022
Prediction of Sweetness using Machine Learning Models Master's project poster presentation	DEC 2018

Academic Visits and Training

University of Tübingen <i>Visiting Doctoral Researcher</i> OpenNuXL integration and validation Integrated and tested an updated OpenNuXL implementation in Proteome Discoverer 3.1.	Nov 2022 Tübingen, Germany
Helix BioGenesis & DNA Xperts <i>Trainee</i> RNA-seq and transcriptomic data analysis Built de novo and reference-based RNA-seq workflows from Illumina data; gained wet-lab experience.	Nov 2016 - MAY 2017 Noida, India

Technical Skills

Programming: Python (scikit-learn, matplotlib, PyTorch), C#, Git; R; exposure to C, SQL, HTML, PHP.
Software & Workflow: Proteome Discoverer 3.1, NuXL, Xcalibur, standard NGS tools, Docker, Linux (Debian/CentOS), Slurm, HPC.

Selected Courses and Training

Machine Learning for Biomedical Applications; Algorithms in Computational Biology; Machine learning in Python with scikit-learn; Basics of large language models - transformers to LLMs (EMBL-EBI).

Awards and Honours

Marie Skłodowska-Curie Actions Fellowship - 2021	Thermo Fisher Scientific, Bremen
Government Fellowship (M.Tech.) - 2017	IIIT-Delhi, India
Bioinformatics Training Fellowship - 2016	Biotech Consortium India Limited

Academic Service

Teaching Assistant: IIIT-Delhi, July 2017–July 2019.
Led tutorials, evaluated presentations, and prepared and assessed exams and quizzes.

Selected Schools and Conferences

EuBIC-MS Winter School - 2024	Winterberg, Germany
PROTrEIN Winter School - 2023	Ghent, Belgium
Mindtrek Conference - 2023	Tampere, Finland
Summer School on Computational Mass Spectrometry - 2022	Barcelona, Spain
PROTrEIN Winter School - 2022	Online
PROTrEIN Summer School - 2021	Online

References

Dr. Mathias Wilhelm
Former PhD Supervisor
Professor
Technical University of Munich
(TUM)
Freising, Germany
mathias.wilhelm@tum.de
+49-8161712701

Dr. Ganesh Bagler
Master Thesis Advisor
Professor
Indraprastha Institute of
Information Technology Delhi
(IIIT-D)
New Delhi, India
bagler@iiitd.ac.in
+91-7793820447

Dr. Vibhor Kumar
Course Instructor
Associate Professor
Indraprastha Institute of
Information Technology Delhi
(IIIT-D)
New Delhi, India
vibhor@iiitd.ac.in
+91-9205424866